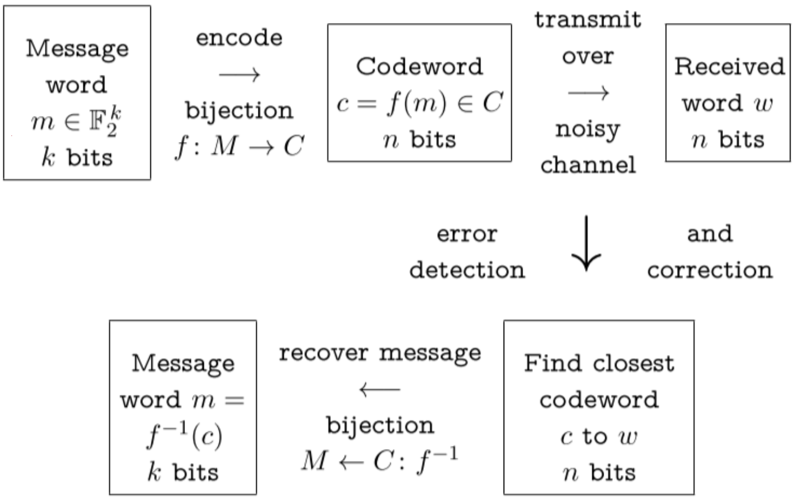
Summary from Monday:



k - length of message words (dimension) $\quad M = \mathbb{F}_2^k$

n - length of codewords $\quad C \subseteq \mathbb{F}_2^n \quad |C| = 2^k$

δ - min. distance between any 2 codewords

An (n,k,δ)-code is $(\delta-1)$-error detecting.

An (n,k,δ)-code is $\left\lfloor \frac{\delta-1}{2} \right\rfloor$-error correcting.

An (n,k,δ)-code has rate $\frac{k}{n}$.

An (n,k,δ)-code has relative distance $\frac{\delta}{n}$.

For a binary symmetric channel with reliability p, $\mathbb{P}_p(c \to w) = p^{n-d(c,w)}(1-p)^{d(c,w)}$ is the probability that w is received if c is transmitted

Exercise: Suppose that 10^6 bits per second are transmitted over a BSC with reliability $p = 0.999999$. What is the expected frequency of words with undetected errors if:

(a) The trivial code of all words in $\mathbb{F}_2^8$ is used?

(b) The parity check code of dimension 8 is used, that is, the code consisting of all even weight words in $\mathbb{F}_2^9$?

(a) Every error is undetected.

P(undetected error in any word)

$= 1 - P(\text{no error in word})$

$= 1 - (0.999999)^8$

$= 1 - (1 - 10^{-6})^8$

$\approx 1 - (1 - 8 \cdot 10^{-6})$

$= 8 \times 10^{-6}$

useful approximation

$(1+x)^n \approx 1 + nx$ for small $|x|$.

Since

$(1+x)^n = \binom{n}{0}x^0 + \binom{n}{1}x^1 + \underbrace{\binom{n}{2}x^2 + \cdots \binom{n}{n}x^n}_{\text{small}}$

$\approx 1 + nx$

Transmission rate:

$$10^6 \frac{\text{bits}}{\text{second}} \cdot \frac{1 \text{ word}}{8 \text{ bits}} = \frac{10^6}{8} \text{ words/second}$$

undetected incorrect words per second $= \dfrac{10^6}{8} \times 8 \times 10^{-6} = 1$

(b)

3.5.4 Theorem Suppose communication is via a BSC with reliability p with $0.5 < p < 1$. Let c_1 and c_2 be codewords of length n, and let w be any word of length n. Then

$$\mathbb{P}_p(c_1 \to w) \ \geq\ \mathbb{P}_p(c_2 \to w) \qquad \Longleftrightarrow \qquad d(c_1,\, w) \leq d(c_2,\, w).$$

In other words, the closer a codeword is to w, the more likely it is that it was the word sent.

Proof First note that $0.5 < p < 1 \implies 0 < 1 - p < p$. Hence $\dfrac{p}{1-p} > 1$. Let $d_1 = d(w,\, c_1)$, $d_2 = d(w,\, c_2)$. Then:

$$\begin{array}{rccll}
 & \mathbb{P}_p(c_1 \to w) & \geq & \mathbb{P}_p(c_2 \to w) & \\
 & \| & & \| & \\
\Longleftrightarrow & p^{n-d_1}(1-p)^{d_1} & \geq & p^{n-d_2}(1-p)^{d_2} & \text{by Equation (3.5.2)} \\
\Longleftrightarrow & \left(\dfrac{1-p}{p}\right)^{d_1} & \geq & \left(\dfrac{1-p}{p}\right)^{d_2} & \\
\Longleftrightarrow & \left(\dfrac{1-p}{p}\right)^{d_1-d_2} & \geq & 1 & \\
 & \| & & & \\
\Longleftrightarrow & \left(\dfrac{p}{1-p}\right)^{d_2-d_1} & \geq & 1 & \\
\Longleftrightarrow & d_2 & \geq & d_1 & \text{because } \dfrac{p}{1-p} > 1.
\end{array}$$

$\square$

Since $d(c_1, w) = \|c_1 + w\|$, we can restate Theorem 3.5.4 as

$$\mathbb{P}_p(c_1 \to w) \geq \mathbb{P}_p(c_2 \to w) \iff \|c_1 + w\| \leq \|c_2 + w\|.$$

The most likely codeword sent is the one with the error pattern of smallest weight.

Let C be a code transmitted over a BSC. Let $c_0 \in C$.

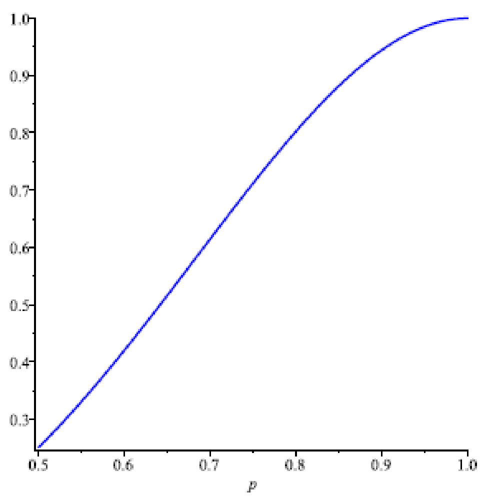
- ★ Let $Q_C(c_0)$ denote the probability that if c_0 was sent, the received word will be successfully corrected back to c_0.
- ★ Let Q_C be the average[3] of all the $Q_C(c_0)$ as c_0 varies across C. This is a measure of the overall success rate of C.

The overall *failure probability* of the code is

$$F_C = 1 - Q_C.$$

Example (in notes). Determine Q_C for the 3-fold repetition code $C = \{000000, 010101, 101010, 111111\}$.

Start by determining $Q_C(000000)$.



<u>Exercise</u> Consider the code

$$D = \{0000, 1000, 0111, 1111\}.$$

(a) Determine Q_D in terms of the channel reliability p.

(b) What is the minimum reliability p of the channel that would provide an overall failure rate F_D of less than 0.01?

3.7. Shannon's Theorem

Suppose we have a BSC with reliability $p > 0.5$. We would like to be able to produce codes that can be used to transmit information across the channel such that the failure probability F_C can be made arbitrarily small. More precisely, we would like a family of codes C_1, C_2, of blocklengths $n_1 < n_2 < \cdots$ such that the failure probabilities $F_{C_1}, \ F_{C_2}, \ \ldots \ \to 0$ as $n \to \infty$. This allows us to pick a suitable code to achieve whatever accuracy is required.

Of course, this is possible by taking n huge compared to k, so there is a gigantic amount of redundancy and extremely low rate $r \to 0$ as $n \to \infty$ .... This is not very interesting.

Instead, we would like to know that we can achieve such a family of codes with r staying fixed as $n \to \infty$. So that, with only a constant percentage overhead, we can achieve failure probability arbitrarily close to 0. It is not obvious that this is possible.

Shannon's Theorem says that it is possible, provided r is not required to be too large. We say that the *channel capacity* R of the BSC is the largest rate for which it is possible. More precisely, the *channel capacity* R of the BSC is the supremum of all rates r for which there exists a family of codes C_1, C_2, ... all of rate $\geq r$, of strictly increasing blocklengths, such that $F_{C_n} \to 0$ as $n \to \infty$.

Roughly speaking, we can transmit information at a rate slightly less than R across our BSC, with essentially no errors (we can arrange that the probability F_C of failure is $< \epsilon$). But with rate slightly above R we cannot achieve this goal, no matter how large we make n. So R is the intrinsic limit of what is possible to achieve with this channel.

3.7.1 Theorem [Shannon, 1948] The channel capacity of a BSC with reliability p exists, and is given by

$$R = 1 + p\log_2(p) + (1-p)\log_2(1-p).$$

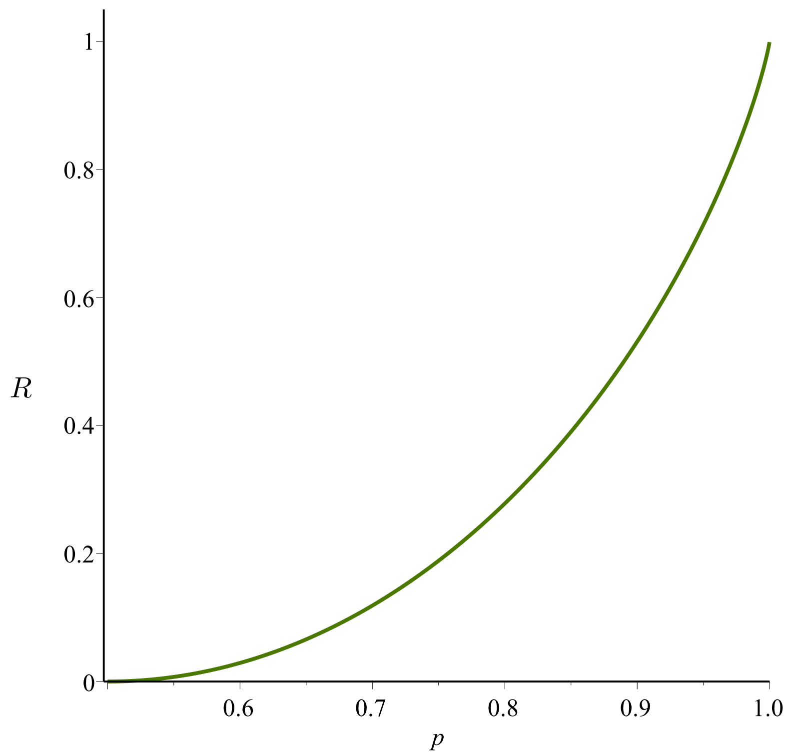
<u>Example</u> Suppose we have a BSC with reliability $p = 0.99$.

Shannon's Theorem says the channel capacity is

$$R = 1 + 0.99\log_2(0.99) + (0.01)\log_2(0.01)$$

$$\approx 0.9192$$

Below is a plot of the channel capacity R of a BSC with reliability p, as given by Shannon's Theorem.



Reliability p	0.50	0.66	0.75	0.80	0.85	0.90	0.95	0.98	0.99	0.999
Capacity R	0	0.075	0.189	0.278	0.390	0.531	0.714	0.859	0.919	0.989

CHAPTER 4

Linear Codes I

If we translate all the words in C by a fixed word, then the distance between codewords, and hence the characteristics of C do not change. Thus wlog we may assume $\mathbf{0} \in C$. Now recall Lemma 3.2.8, which says $(C+C)\setminus\{0\}$ is the set of non-zero error patterns that C cannot detect. We would like to minimize this set.

Since $\mathbf{0} \in C$, $\mathbf{0} + c = c \in C + C$ for every $c \in C$, so $C + C \supseteq C$. So the best (smallest) we could hope for is $C + C = C$, which says C should be *closed under addition*. This means that C will be a *vector space* over $\mathbb{F}_2$, and $f\colon \mathbb{F}_2^k \to \mathbb{F}_2^n$ should be an injective map between vector spaces. The most natural choice for f would be a *linear map*. Then f will be described by a matrix of full rank.

4.1.1 Definition A code C is a *linear code* if the sum of any two codewords is also a codeword. That is if $v, w \in C$, then $v + w \in C$. We shall assume $C \neq \{\mathbf{0}\}$, ie that $k > 0$.

4.1. Review of Linear Algebra

We can do linear algebra over $\mathbb{F}_2$ in the same way as over $\mathbb{R}$ or $\mathbb{C}$. All of the same methods and results still apply: row reduction, determinants etc.

Recall that a non-empty subset $U \subseteq \mathbb{F}_2^n$ is a $\mathbb{F}_2$-*vector space* iff U is closed under addition and scalar multiplication. Since the only scalars are $0, 1 \in \mathbb{F}_2$, and $0 \cdot v = \mathbf{0}$, U is a vector space iff $\left[\mathbf{0} \in U \text{ and } U \text{ is closed under addition}\right]$. If $U \subseteq \mathbb{F}_2^n$ is non-empty and closed under addition, then U must contain the zero vector (since if $u \in U$, then $u + u = \mathbf{0} \in U$), so will be an $\mathbb{F}_2$-vector space.

Thus a linear code is exactly a vector subspace of $\mathbb{F}_2^n$.

A *linear combination* of vectors $v_1, \ldots, v_n$ is an expression $\sum \alpha_i v_i$, where the α_i are scalars. Over $\mathbb{F}_2$ the only scalars are 0 and 1, so a linear combination of vectors is just a sum of some of them: v_4, $v_1 + v_3 + v_5$, $v_1 + \cdots + v_n$, etc.

A set of vectors $\{v_1, \ldots, v_n\}$ is *linearly independent* if no linear combination of them is equal to $\mathbf{0}$, except the trivial combination (empty sum) $0v_1 + \cdots + 0v_n$. Since the only scalars are 0 and 1, $S = \{v_1, \ldots, v_n\}$ is linearly independent iff no non-empty sum of v_i adds to $\mathbf{0}$. If a set of vectors is not linearly independent, then the set of vectors is said to be *linearly dependent*.

Exercise: Let u, v, w be non-zero vectors in $\mathbb{F}_2^n$.

(a) When is $\{u, v\}$ linearly dependent?

(b) For distinct u, v, w, when is $\{u, v, w\}$ linearly dependent?

(c) Find a set of four linearly dependent vectors such that every proper subset of the vectors is linearly independent.

The *span* of $S = \{v_1, \ldots, v_n\}$, denoted $\operatorname{span}(S)$ or $\langle S \rangle$ is the collection of all linear combinations that can be made from S. This is always a vector space.

A *basis* for a vector space V is a subset $B \subseteq V$ such that:

★ $\langle B \rangle = V$ (B spans V), and

★ B is linearly independent.

Every vector space V has a basis. In fact, a vector space usually has many different bases, but all bases of V have the same number of elements. The *dimension* of a vector space is the number of elements in (any) basis.

Exercise Let $S = \{1101, 1001, 0101, 0100\}$.

(a) Determine $\langle S \rangle$.

(b) Find a basis for $\langle S \rangle$.

We also consider matrices with entries in $\mathbb{F}_2$.

Abbreviations:

- ★ ero = elementary row operation.
- ★ ref = row echelon form of a matrix.
- ★ rref = reduced row echelon form (after Gauss-Jordan reduction).

The only eros are: adding two rows, and swapping two rows.

4.1.3 Theorem Let $\mathbf{A}$ be an $m \times n$ matrix with entries in $\mathbb{F}_2$. All of the following quantities are then equal:

- ★ The number of linearly independent rows in $\mathbf{A}$
- ★ The dimension of the space spanned by the rows of $\mathbf{A}$ (row space)
- ★ The number of linearly independent columns in $\mathbf{A}$
- ★ The dimension of the space spanned by the columns of $\mathbf{A}$ (column space)
- ★ The number of non-zero rows in any ref of $\mathbf{A}$.

Exercise Find the rref of $\begin{pmatrix} 1 & 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 & 0 \end{pmatrix}$ in $\mathbb{F}_2$.

4.1.4 Definition The number of linearly independent rows of a matrix $\mathbf{A}$ is called the *rank* of $\mathbf{A}$. This is equal to the other quantities listed in Theorem 4.1.3. The set of all v such that $\mathbf{A}v = \mathbf{0}$ is called the *nullspace* or *kernel* of $\mathbf{A}$, and denoted $\mathrm{NS}(\mathbf{A})$.

4.1.5 Theorem Let $\mathbf{A}$ be an $m \times n$ matrix with entries in $\mathbb{F}_2$. Then

$$\operatorname{rank}(\mathbf{A}) + \dim \mathrm{NS}(\mathbf{A}) = n.$$

One of the advantages of a linear code is that its distance is easy to find.

4.2.1 Theorem For any linear code $C \neq \{\mathbf{0}\}$, the distance of C is equal to the smallest weight of any nonzero codeword.

Proof Let C be a linear code of distance δ, and let c_0 be a nonzero codeword of smallest weight. Since $\mathbf{0}$, $c_0 \in C$ the distance of C is at most $d(c_0, \mathbf{0}) = \|c_0\|$, so $\delta \leq \|c_0\|$. Suppose $\delta < \|c_0\|$. Then there are two codewords c_1, $c_2 \in C$ such that $\delta = d(c_1, c_2) < \|c_0\|$. As C is linear, $c_1 + c_2$ must also be a codeword, of weight $\|c_1 + c_2\| = d(c_1, c_2) < \|c_0\|$. But this contradicts the assumption that c_0 is the nonzero codeword of smallest weight. $\square$

It is easy to find the distance of a linear code. Other advantages of linear codes include:

- ★ For linear codes, there is a procedure for IMLD which is simpler and faster than we have seen so far (for some linear codes, there are very simple decoding algorithms).
- ★ Encoding using a linear code is faster than for arbitrary non-linear codes.
- ★ It is easy to manufacture linear codes. Just take the span of any set of vectors.
- ★ It is easy to describe the set of error patterns that a linear code will detect.
- ★ It is much easier to describe the set of error patterns a linear code will correct than it is for arbitrary non-linear codes.

Disadvantage: linear codes are not necessarily optimal in terms of their error correcting ability.

Recall that an (n, k, δ) code has 2^k codewords of length n, and minimum distance δ.

A *basis for a linear code* C is a basis for the vector space C. The *dimension of* C is its dimension as a subspace of $\mathbb{F}_2^n$, namely the number of elements in a basis.

For a linear code we have now given two different definitions of dimension. However they agree: Suppose a linear code C has dimension k in the vector space sense. Let $B = \{c_1, c_2, \ldots, c_k\}$ be a basis for C. Then each codeword b in C can be written as

$$b = a_1c_1 + a_2c_2 + \cdots + a_kc_k$$

for a unique choice of scalars a_1, a_2, $\ldots$, a_k. Since each a_i is 0 or 1 for $1 \leq i \leq k$, there are 2^k distinct choices for a_1, a_2, $\ldots$, a_k, and hence for b.

So the dimension of C in the linear algebra sense agrees with our definition of the dimension k of a code in the general case.

How can we describe a vector space C? Two approaches are: give a basis, or give a set of equations defining the elements of C.

If c_1, ..., c_k forms a basis of C, then every element $b \in C$ can be written uniquely in the form $b = a_1 c_1 + \cdots + a_k c_k$. Let **G** be the matrix formed by taking row i to be c_i for $1 \leq i \leq k$. Then expressions $a_1 c_1 + \cdots + a_k c_k$ are exactly the vectors obtained by multiplying **G** by the vector $(a_1, a_2, \ldots, a_k)$. That is, C is the image of the matrix **G**.

Alternatively, one can describe C as all vectors satisfying some linear equations, such as $C = \{x \mid x_1 + x_3 = 0,\ x_2 + x_4 = 0\}$ etc, where x is the word with bits $x_1 x_2 \cdots$. This says x is an element of the nullspace of some matrix **H**.

That is, C may be defined as the image of one map (**G**) or the kernel of another (**H**).

In detail, we shall show that associated to each linear code C are two matrices, denoted **G** and **H**:

- ★ **G** is used for encoding.
- ★ **H** is used for error correction/detection and for finding the distance δ.

First, recall the method of finding a basis for a linear code (vector space):

4.3.1 Algorithm [Algorithm for finding a basis for $C = \langle S \rangle$.]

Let S be a non-empty subset of $\mathbb{F}_2^n$.

- ★ Form the matrix $\mathbf{A}$ whose rows are the words in S.
- ★ Use eros to find a ref (or rref) of $\mathbf{A}$.
- ★ The non-zero rows of the ref of $\mathbf{A}$ form a basis for $C = \langle S \rangle$.

<u>Exercise</u>: Find a basis for the linear code $C = \langle S \rangle$ where $S = \{0110, 1100, 1001, 0101\}$.

4.4. Generating Matrices

We now use the material from the previous section to create a matrix that is used in the encoding process for a linear code.

4.4.1 Definition If C is a linear code of length n and dimension k, then any matrix whose rows form a basis for C is called a *generating matrix* for C. Such matrices must have k rows and n columns, and have rank k.

Since a vector space may have many different bases, the matrix $\mathbf{G}$ is not unique. (It is unique if we insist that it is in rref.)

4.4.2 Theorem A $k \times n$ matrix $\mathbf{G}$ with $n \geq k$ is a generating matrix for some (n, k) linear code C iff the rows of $\mathbf{G}$ are linearly independent (so the rank of $\mathbf{G} = k$).

4.4.3 Theorem If $\mathbf{G}$ is a generating matrix for a linear code C, then any matrix row equivalent to $\mathbf{G}$ is also a generating matrix for C. In particular, every linear code has a unique generating matrix in rref.

4.4.5 Example Let C be the linear code $C = \{000000,\ 001110,\ 010101,\ 011011,\ 100011,\ 101101,\ 110110,\ 111000\}$. Show that $\mathbf{G}$ below is a generating matrix for C.

$$\mathbf{G} = \begin{pmatrix} 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 \end{pmatrix}.$$

Solution: Use Algorithm 4.3.1. Or note that

- ★ $|C| = 8$ so $\dim C = 3$, and $\mathbf{G}$ is 3×6,
- ★ The rows of $\mathbf{G}$ are linearly independent,
- ★ Each individual row of $\mathbf{G}$ is a codeword.

So the rows of $\mathbf{G}$ generate a 3-dimensional subspace of the 3-dimensional space C, and hence must exactly generate C.

4.4.6 Theorem If $\mathbf{G}$ is a generating matrix for a linear (n, k)-code C, then $c = m\mathbf{G}$ ranges over all 2^k codewords in C as m ranges over all 2^k words of length k. Thus $C = \{m\mathbf{G} \mid m \in \mathbb{F}_2^k\}$. Moreover, $m_1\mathbf{G} = m_2\mathbf{G}$ iff $m_1 = m_2$.

Message m is encoded as $c = m\mathbf{G}$, where m and c are represented by row vectors.

Example: Using $G = \begin{pmatrix} 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 \end{pmatrix}$ for code C from Example 4.4.5, encode the message word 011.

$G' = \begin{pmatrix} 1 & 0 & 1 & 1 & 0 & 1 \\ 1 & 1 & 0 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 \end{pmatrix}$ is another generating matrix for C.

Encode the message word 011 using G^*.

4.5.1 Definition Let $\mathbf{G}$ be a $k \times n$ matrix with $k \leq n$. We say $\mathbf{G}$ is in *standard form* if the first k columns of $\mathbf{G}$ form the $k \times k$ identity matrix $\mathbf{I}_k$. That is, $\mathbf{G} = (\mathbf{I}_k \mid \mathbf{X})$. Any such $\mathbf{G}$ automatically has linearly independent rows and is in rref. Thus $\mathbf{G}$ is a generating matrix for some linear code of length n and dimension k. We say that the code C generated by $\mathbf{G}$ is *systematic.*

Not all linear codes have a generating matrix in standard form.

4.5.2 Example Is the linear code $C = \{00000,\ 10110,\ 10101,\ 00011\}$ systematic?

Solution: We find a generating matrix. There is no need to write down the $\mathbf{0}$ word. ($\mathbf{0}$ is never part of a basis.) So:

$$\begin{pmatrix} 1 & 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{pmatrix} \to \begin{pmatrix} 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{pmatrix} \to$$

$$\begin{pmatrix} 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \to \begin{pmatrix} 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \quad \text{rref}$$

So $\mathbf{G} = \left(\begin{array}{cc|ccc} 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{array}\right)$. The unique rref is not in standard form, so C is not systematic. ⊛

4.5.3 Theorem If C is a linear code of length n and dimension k with generating matrix $\mathbf{G}$ in standard form and a message m is to be transmitted, then the first k bits of the codeword $v = m\mathbf{G}$ form the message word $m \in \mathbb{F}_2^k$.

4.5.5 Definition If C is any code of length n, we can obtain a new code C' of length n by applying a fixed permutation σ of the n bits. The resulting code $C' = \sigma(C)$ is said to be *equivalent* to C.

4.5.7 Theorem Any linear code C is equivalent to a linear code having a generating matrix in standard form.

Exercise Consider the (6,3) linear code with generating matrix $G = \begin{pmatrix} 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{pmatrix}$

(a) Determine the codeword corresponding to the message word 011.

(b) Determine the message word corresponding to the codeword 101101.

(c) What is the distance of this code?